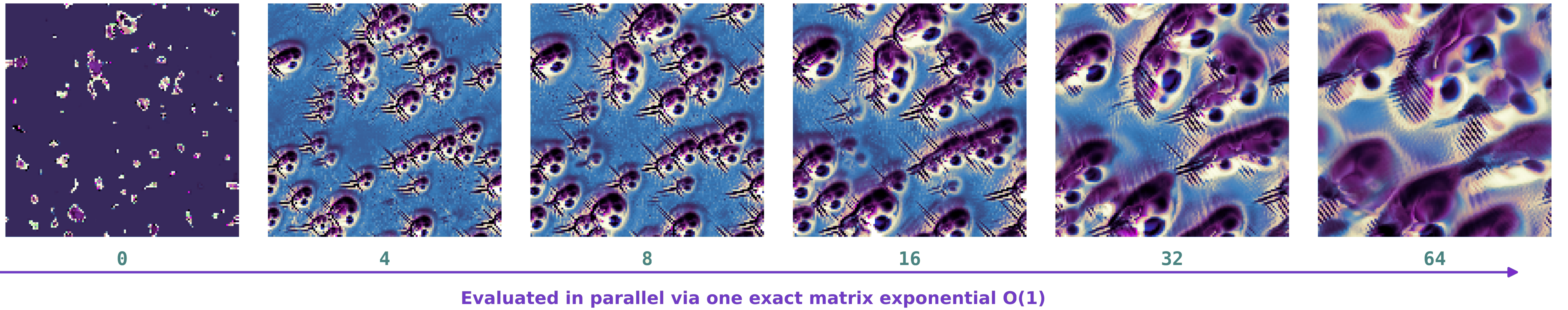


Closed-Form Neural Cellular Automata

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The Bottleneck

Standard NCAs require $\mathcal{O}(T)$ sequential steps, creating a severe memory bottleneck during Backpropagation Through Time (BPTT). Our work enables gradient-based design of target patterns without the memory cost of simulating 1000+ steps. While demonstrated on CLIP feature visualization, this approach can model any continuous spatial signal.

$$\mathbf{x}(t) = \mathbf{P} \left(\mathcal{F}^{-1} \left(\mathbf{U} \cdot \exp(t \lambda(\xi)) \cdot \mathbf{U}^T \cdot \mathcal{F}(\mathbf{L}(\mathbf{x}_0)) \right) \right)$$

$\mathbf{x}_0, \mathbf{x}(t)$: Input spatial noise seed and final image.

$\mathbf{L}, \mathbf{P}$: Deep nonlinear Lift and Project neural networks.

$\mathcal{F}, \mathcal{F}^{-1}$: Spatial Fourier transform and its inverse.

The Constant-Time Solution

By bounding strictly linear hidden dynamics between nonlinear encoders, we diagonalise the transition matrix in the Fourier domain. The T -step rollout mathematically collapses into a single exact equation, reducing temporal complexity to $\mathcal{O}(1)$ in T and granting instant random access to any time t .

$\mathbf{U}, \mathbf{U}^T$: Orthogonal matrices mixing hidden channels.

$\lambda(\xi)$: Fourier spectrum of the learned spatial kernel.

$\mathcal{F}(\mathbf{L}(\mathbf{x}_0))$: Spatial FFT of the initial hidden state $\hat{\mathbf{h}}_0$.

Scope & Limitations

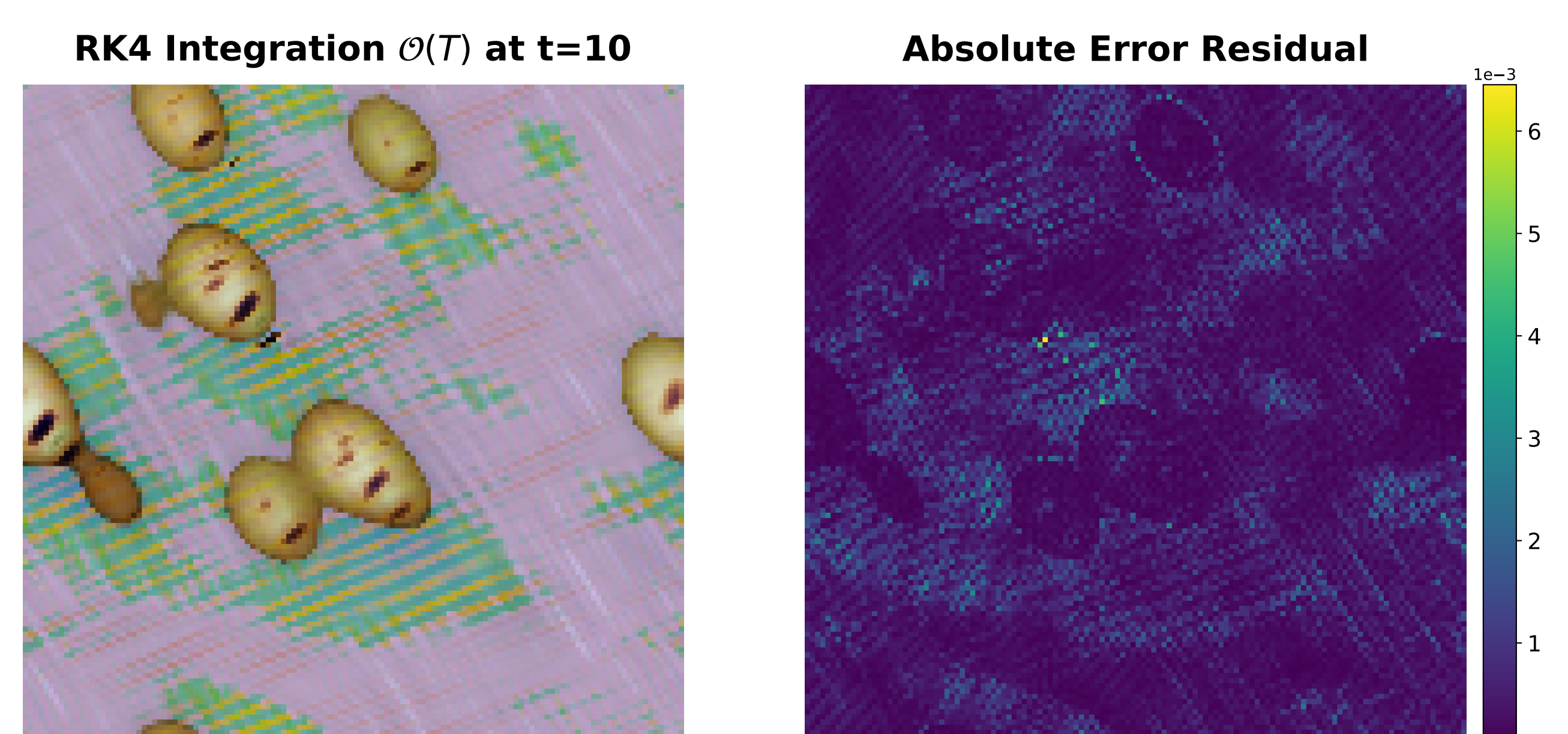
Linear Core Restriction:

Strict linearity in the hidden dynamics restricts highly nonlinear emergence (e.g., Game of Life, Lennia), focusing instead on stable, parameterised fluid pattern generation.

Spatial vs. Temporal Cost:

Spatial complexity is $\mathcal{O}(N \log N)$ due to the FFT, but temporal complexity is strictly $\mathcal{O}(1)$ in T . This trades spatial compute for infinite-horizon temporal evaluation.

Verification



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Code: github.com/algroznykh/closed_form_nca